

Name: _____

SOLUTIONS

Date: _____



Year 9 Maths

Investigation 2, 2012

Topic – Space and Measurement

Take - Home Component

$$\frac{\quad}{52}$$

$$= \quad \%$$

Weighting

3% of the year.

BONUS MARKS (look at the whole assessment):

☐ Units: appropriate (+1). If inconsistent (+1/2 only)

Equipment:

Calculator

Important Information:

This part of the assignment will be completed at home over a period of up to one week. This part of the assignment is worth 3% of the marks for the year.

AIM: *In this assessment, you will be investigating the perimeter of figures made from 1cm squares..*

Marks Summary

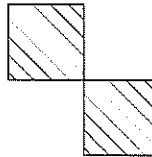
Task	Marks available	Marks awarded
1	-	-
2	3	
3	10	
4	12	
5	16	
6	8	
7	2	
Bonus	1	
Part 1	52	

Task 1 [No marks]

- a) This is a one centimetre (1 cm) square. That means it has sides of 1 cm.
- The perimeter is the distance around the outline of the shape and is therefore 4 cm.



- b) Here are two squares that meet at a corner.
- The perimeter of this shape is 8 cm.



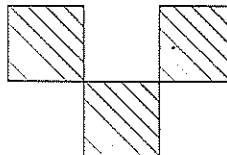
- c) Here are two squares that share a side, with the side they share shown in bold.
- The distance around the outside of the shape (its perimeter) is 6 cm.
- You will notice that sharing sides reduces the perimeter.



Task 2 [3 Marks]

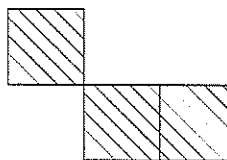
These diagrams show examples of three connected 1 cm squares with various numbers of shared sides. Calculate the perimeter of each.

- a) No shared sides



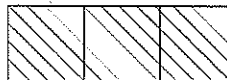
Perimeter = 12 cm [1]

- b) One shared side



Perimeter = 10 cm [1]

- c) Two shared sides

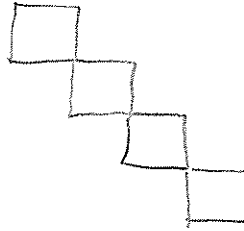


Perimeter = 8 cm [1]

Task 3 [10 Marks]

Draw diagrams to show examples of four connected 1cm squares with various numbers of shared sides and calculate the perimeter of each.

a) No shared sides



Perimeter = 16 cm [1]

[1]

b) One shared side



Perimeter = 14 cm [1]

[1]

c) Two shared sides



Perimeter = 12 cm [1]

[1]

d) Three shared sides



Perimeter = 10 cm [1]

[1]

e) Four shared sides



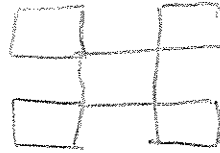
Perimeter = 8 cm [1]

[1]

Task 4 [12 Marks]

Draw diagrams to show examples of five connected 1cm squares with various numbers of shared sides and calculate the perimeter of each.

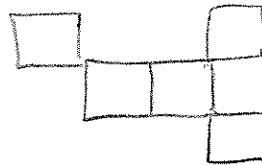
a) No shared sides



Perimeter = 20 cm [1]

[1]

b) One shared side



Perimeter = 18 cm [1]

[1]

c) Two shared sides



Perimeter = 16 cm [1]

[1]

d) Three shared sides



Perimeter = 14 cm [1]

[1]

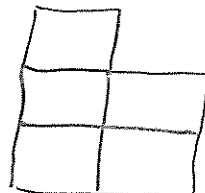
e) Four shared sides



Perimeter = 12 cm [1]

[1]

f) Five shared sides



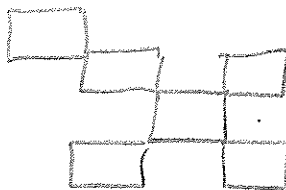
Perimeter = 10 cm [1]

[1]

Task 5 [16 Marks]

Draw diagrams to show examples of six connected 1cm squares with various numbers of shared sides and calculate the perimeter of each.

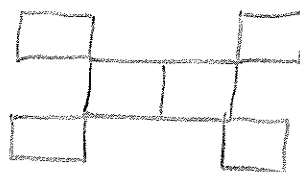
a) No shared sides



Perimeter = 24 cm [1]

[1]

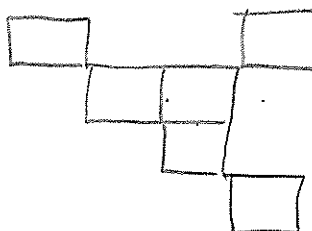
b) One shared side



Perimeter = 22 cm [1]

[1]

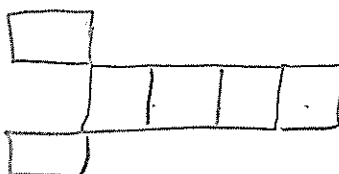
b) Two shared sides



Perimeter = 20 cm [1]

[1]

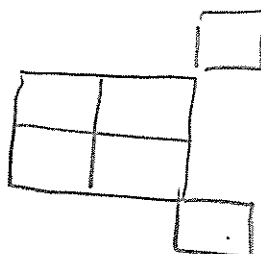
c) Three shared sides



Perimeter = 18 cm [1]

[1]

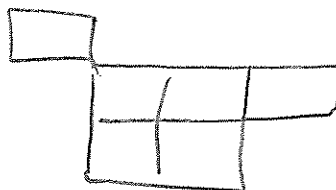
d) Four shared sides



Perimeter = 16 cm [1]

[1]

e) Five shared sides



Perimeter = 14 cm [1]

[1]

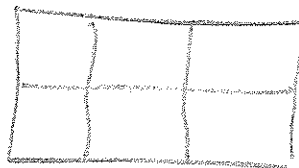
f) Six shared sides



Perimeter = 12 cm [1]

[1]

g) Seven shared sides



Perimeter = 10 cm [1]

[1]

Task 6:[8 Marks: 2, 2, 2,2]

Summarise your findings by putting the results from the above tasks into the table below:

Number of squares	Maximum perimeter (cm)	Minimum perimeter (cm)
1 (Task 1)	4	4
2 (Task 1)	8	6
3 (Task 2)	12	8
4 (Task 3)	16	8
5 (Task 4)	20	10
6 (Task 5)	24	10

Task 7: [2 Marks]

Do you notice any connection between the number of squares and

- the maximum perimeter?
- the minimum perimeter?

If so make notes here:

MAXIMUM PERIMETER = 4 X NUMBER OF SQUARES

MINIMUM PERIMETER = MAXIMUM - 2 X SHARED SIDES
 MAXIMUM NUMBER OF SIDES